6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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Abstract.

Understanding climate change relies heavily on grasping the role of cloud radiative effect and its climate feedback. However, these factors pose the greatest uncertainty among atmospheric processes. This uncertainty stems partly from the limited global-scale cloud measurements. To address this gap, our study presents a six-year statistical analysis of cloud coverage and microphysics properties at the ACTRIS-CCRES AGORA observatory in Granada, Spain. Here, we utilized CLOUDNET post-processed data (e.g target classification, liquid water path, droplet effective radius and so on) of remote sensing instruments and microwave radiometer thermodynamic profiles. The analysis focuses on single layer cloud properties comparison for liquid, ice, mixed phase, liquid with rain, and mixed phase with rain clouds. Therefore, a distinct seasonal pattern in cloud occurrence was observed. During the summer (June-August), single layer clouds occurrence is less than 10%, and precipitation is absent. Conversely, cloud occurrence significantly increases during winter (December-February) and spring (March-May), where Ice and mixed phase clouds with rain are the predominant type. Additionally, a negligible presence of liquid clouds with rain were found. Seasonal trends in liquid water content (LWC) and droplet effective radius are unclear for mixed-phase and liquid clouds. Effective radius remains stable year-round, around 5 μm for liquid clouds below 3 km, while for mixed-phase clouds it can vary between 5 and 12 μm, maily found at 1-2 km and 5-9 km. LWC shows higher values in cold periods and lower in summer, but with significant variability across both cloud types. Same seasonal patterns for ice water content (IWC) in ice and mixed-phase clouds were challenging to identify. Monthly profiles showed little variation, although values tended to rise in summer. Ice clouds typically have effective radius between 20-30 μm at 8-12 km altitude, peaking at 65 μm in summer. Mixed-phase clouds exhibit ice effective radius around 50 μm between 1-7 km altitude. The mean geometric thickness of ice and mixed-phase clouds throughout the seasons is 1.2 $\pm$ 1.3 km and 2.0 $\pm$ 1.6 km, respectively, contrasting sharply with the 0.2 $\pm$ 0.2 km observed in liquid clouds. The monthly variability in thickness is substantial enough to obscure any seasonal differences. Similarly, the analysis of cloud base height reveals mean values of 1.9 $\pm$ 2.0 km (excluding summer) for liquid clouds, 4.8 $\pm$ 2.2 km for mixed-phase clouds, and 7.0 $\pm$ 2.4 km for ice clouds. During summer a different vertical distribution of cloud base was found due to different mechanisms of cloud formation. Conclusively, a statistical analysis was conducted on single-layer cloud properties in the Southeast of Spain, highlighting the maily cloud type associated with rainfall in the region.

1 Introduction

Clouds are one of the major atmospheric components driving the earth's radiative budget and hydrological cycle variability. Since they roughly cover 2/3 of the Earth's surface **?**, their physical properties play an important role in climate feedbacks such as surface temperature change rate. In particular, clouds are responsible for precipitation patterns and can contribute to increased surface temperature by trapping thermal radiation in the atmosphere **?**. Given that, raised temperatures and water shortage accounts for more severe drought stress in mediterranean regions **??**. In addition, the fifth assessment report **?** suggests the Mediterranean basin will suffer a reduction of precipitation **??**. In general, this area is known for experiencing frequent severe weather events and significant climate variability **??**, being of particular scientific and societal interest. However, surface temperature feedback due cloud radiative effect is still poorly understood **?**. Moreover, further projections using GCMs (global climate models) are still challenging due to the difficulty in partitioning liquid and ice content for mixed phase clouds **?**. Particularly, assimilation of cloud types with precipitation showed the importance of ice process in driving precipitation for warm clouds **??**. From this perspective, measurements of cloud doppler radar can be used to retrieve droplets and ice microphysics properties such as effective radius, liquid water (LWC) content and ice water content (IWC) **???**. In combination with a ceilometer, they can also determine cloud boundaries, i.e. base and tops. Thus, in order to understand seasonal trends on cloud occurrence and their physical properties, long-term measurements of remote sensing instruments are necessary. Therefore, a statistical description of cloud type occurrences in southeast Spain using passive and active remote sensing measurements to comprehend cloud and precipitation patterns in the Mediterranean basin was achieved.

The presented study used more than half a decade of Cloudnet post-processed data **?** and microwave radiometer profiles at the Spanish AGORA-CCRES station (Andalusian Global ObseRvatory of the Atmosphere - Centre of Cloud Remote Sensing). The AGORA station has been a part of the Cloudnet since yearXXX, and was integrated into ACTRIS CCRES in yearXXX as its new facility. Their products have been continuously determined using a synergistic approach of cloud observations using a 94-GHz doppler radar, with multiple scanning capabilities, a microwave radiometer and a CHM15k ceilometer. Here, Cloudnet target classification is employed to cloud type classification as **??**. This product also allowed the partitioning of retrieved droplets and ice microphysical properties within each cloud type. Thus, an inter-annual statistical analysis on cloud occurrence and microphysical properties is presented.

Cloudnet products have been explored in many recent studies. **?** used this dataset at German Alps to analyze cloud variability at a particular high-altitude location. **?** studied data from the Leipzig observatory LACROS (Leipzig Aerosol and Cloud Remote Observations System), to evaluate ice process for mixed phase clouds. **??** focused their study on cloud statistics properties for different types of clouds in the Arctic and the Bucharest-Magurele site, respectively. Differently from the previously cited studies, our station has particularly thermodynamic conditions, with elevated temperatures and less humidity. Besides that, we also described microphysical properties for different cloud types, which can be extremely useful for feeding radiative transfer models in order to estimate cloud radiative effects. **?** and **?** used microphysics estimation from XXX and XXX to perform numerical estimation of cloud radiative effect using the libRadtran tool.

The paper is organized as follows. Sec **??** describes the dataset and the methods used for cloud classification. Sec **??** first describes the cloud correlation with thermodynamic conditions and then evaluates the inter-annual cloud properties variability. Finally, SecXXX presents the concluding remarks and contextualize the main findings within the climate research topic.

2 Data and Methods

2.1 Site and database

An extensive dataset ranging the years 2018 to 2023 was used. In particular, Cloudnet (Illingworth et al., 2007) post-processed high level data and microwave radiometer humidity and temperature profiles. Concerning the 6 years of cloudnet dataset, Figure 1 shows the monthly data availability. These products were inferred by measurements carried out with a set of ground-based remote sensing instruments - managed at the ACTRIS-CCRES Granada site facility (CCRES stands for Centre for Cloud Remote Sensing) at 37°2E, 3.6°S and 680 m. This station is as part of the University of Granada (UGR) and it is integrated into the AGORA observatory. It is located within a broad intramontane depression in the foothills of the tallest mountain massif of the Iberian Peninsula, Sierra Nevada.

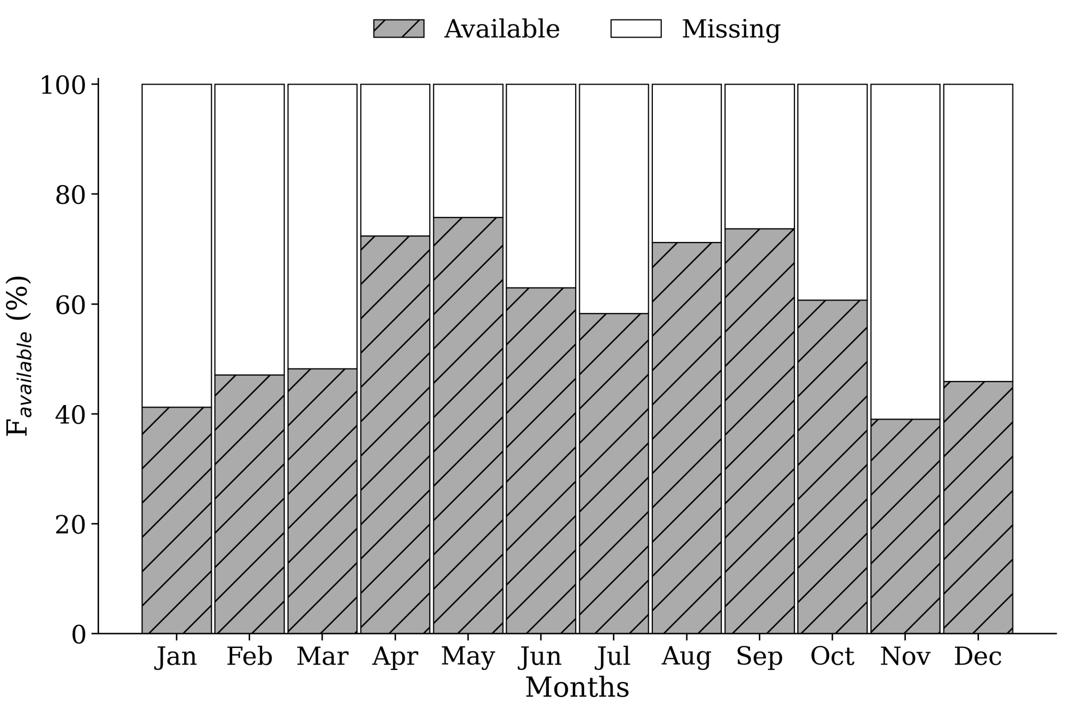


Figure 1. ...

The Cloudnet processed data was determined by a synergy between a RPG-HATPRO radiometer, a CHM15k ceilometer, a RPG-FMCW 94 GHz cloud Doppler radar and the European model ECMWF (an acronym for European Centre for Medium-Range Weather Forecasts). These instruments have been continuously operated since 2018, and their data has been processed by the CloudnetPy (Illingworth et al., 2007), performing reflectivity attenuation correction, error estimation and quality flag assignment. Particularly, attenuation corrections assists in prevention misclassification of atmospheric elements through liquid water content (LWC) estimation. This variable is obtained by cloud boundaries provided by lidar and radar synergy, and microwave radiometer liquid water path (LWP) **?**. Thus, uncertainties retrieving LWC during a raining event can be high enough to frustrate attenuation corrections.

CLOUDNET processed data were sampled in a common grid with time resoltion of 30 s, and height bins according to cloud radar range resolution, which differ according to radar chirp table configuration. This resolution vary with altitude and underwent some changes throughout the measurement period. Consequently, to ensure consistency in vertical resolution of the data, a regridding process was implemented to facilitate monthly statistical analysis when needed. The regriding consisted to change altitude resolution to 60 m. Table 1 associate bin sizes with their respective height intervals. This table also gives operational information of instruments at Granada site and their main post-processed variables. Differently, microwave radiometer humidity and temperature profiles were not processed by CloudnetPy, but were provided by its own manufacturing software. High level data included in the analysis were target classification, liquid water path (LWP) ice water path (IWP), droplet effective radius, and ice effective radius, where target classification apportioned to classify cloud layers as explained in section 2.3. The classification method was adapted from Pîrloagă et al. (2022) and Nomokonova et al. (2019).

Table 1. This is a sample table caption and table layout.

Instrument	**Height resolution (heigh interval) (m)**	**Integration time (s)**	**Variables**
FMCW 94 GHz Cloud Doppler Radar	14.9 (100-1200), 13.5 (1200-4400), 15.5 (4400-10000)	1.10	Mean Doppler vellocity (m/s)
	22.4 (100-924), 25.6 (924-8455), 39.7 (8455-12000)	3.2	Reflectivity (dBz)
	29.8 (100-1200), 29.8 (1200-4900), 29.8 (4900-8500), 51.1 (8500-13500)	3.57	Linear depolarisation ratio (dB)
	29.8 (100-1200), 33.3 (1200-5500), 34.1 (5500-11000)	3.26	Spectral width (m/s)
	36 (180-1476), 12.8 (1476-3998), 12.8 (3998-8158), 12.8 (8158-1200)	2.81	
MWR RPG HATPRO	200 (0-2000), 400 (2000-5000), 800 (5000-10000)	10	Humidity profiles (g m^{-3})
	50 (0-1200), 200 (1200-5000), 400 (5000-10000)	10	Temperature profiles (K)
	–	2	LWP, IWP (kg m^{-2})
CHM15k Nimbus Ceilometer (1064 nm)	15	15	Attenuated backscatter coefficient (sr^{-1} m^{-1})

2.2 Cloudnet particle classification scheme

Target classification product assigns an integer ranging from 0 to 10 at each pixel in the Cloudnet processed grid. Each number are related to atmospheric elements in the following order: clear sky, droplets, drizzle or rain, drizzle & droplets, ice, ice & droplets, melting ice, melting & droplets, aerosol, insects, aerosol & insect. These targets were determined through radar and lidar variables in combination with thermodynamic profiles from the ECMWF model. In particular, wet bulb temperature (T_w) profiles is used to establish warm and cold regions to classify liquid and ice hydrometeors. Further, liquid droplets boundaries were resolved with lidar attenuated backscatter coefficient (β), and radar reflectivity. The first is sensible to small liquid droplets present in cloud base, as the second are able to penetrate the cloud and detect cloud tops. In addition, liquid layers respect a threshold of T_w < -40 °C, since it is not possible to keep water in liquid phase bellow that temperature. Hence, liquid droplets assigned as falling were classified as drizzle or rain droplets, where falling is attributed through radar echos analysis once particles boundary were already determined. Therefore, falling cold particle were just classified as ice, without distinguishing ice from precipitating ice. Next, melting layers were identified at the height of maximum wet bulb temperature divergence. Pixels with divergence surpassing 0.0075 s^{-1} within 150 m of height deviation were also classified as "melting". Furthermore, more than one class of particle can be settled in the same pixel, allowing the identification of mixed phase hydrometeors, like ice coexisting with supercooled liquid droplets and melting ice particles coexisting with cloud liquid droplets. A more detailed description of particle apportionment of Cloudnet algorithm can be found on Hogan and O'connor (2004).

2.3 Assortment algorithm

The assortment algoritm classifies groups of hydrometeors and cloud layers. In terms of hydrometeors, this algorithm generate masks of hydrometeor occurence for 3 groups: liquid, ice and "total", where liquid class encompass all possible liquid pixels, which are "droplets", "drizzle or rain" and "drizzle & droplets". Ice encompass only "ice" pixels, and total is all possible hydrometeors, whic are "droplets", "drizzle or rain", "drizzle & droplets", "ice", "ice & droplets", "melting ice" and "melting & droplets".

Subsequently, cloud layers in atmospheric column were classified trought pixel sequence inquiry. These pixels sequences were devided in the following 5 groups according to Pîrloagă et al. (2022): liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only "ice" pixel sequence was encountered. As well as, mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified wheter at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Diferent from previous studies, a threshold of 20 bins (256-1022 m) bellow cloud was set to reject precipitation events, assuming that rain droplets are not falling from the actual cloud layer. In all cases, for being considered a cloud layer, a minimun sequence of 3 cloud pixels (38-153 m) (Nomokonova et al., 2019) must be satisfied, in addition to a minimun sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Arosol & insect" to separate multilayers. Therefore, free hydrometeor layers in the neghborhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on Pîrloagă et al. (2022); Nomokonova et al. (2019), but theses values are not currently well defined, their determination has been guided by empirical experience.

3 Results and discussion

3.1 Thermodynamic influences on cloud frequency

It is widely recognized that thermodinamic conditions can affect cloud formation (**?**). In order to relate both, this section makes a comparison between monthly thermodinamic profiles and different hydrometeors and cloud type occurence.

Temperature and humidity profiles used as baselines were those collected during clear sky condition, and are shown in Figure 2. This figure can easily identify cold and warm regimes, where cold regimes spans between winter and spring, while warm regimes spans between summer and autumn. This annual variablility was also observed by Pîrloagă et al. (2022); Nomokonova et al. (2019), howevever, higher temperature values experienced by our station is only closely resembling those reported by Pîrloagă et al. (2022), as both stations are geographic located in warm regions. In our case, averange surface temperature is 25.2 $\pm$ 5.5 °C in summer, by contrast with 17.4 $\pm$ 6.5 °C in winter. For monthly averange surface relative humidity, values appear slightly higher than those documented by Nomokonova et al. (2019), with 50 $\pm$ 21% in summer and 64 $\pm$ 17% in winter.

Following this, hydrometeor formation and consequently, cloud formation will be directly affected by thermodinamic conditions in these regimes. For tracking this changes on clouds, a first approach on monthly hydrometeors occurrence is depicted in Figure 3. Regarding the 6-year CLOUDNET post-processed database, this figure shows the monthly frequency profiles of liquid (top panel) and ice (middle panel) hydrometeors. The lowest panel also displays their corresponding monthly frequency of occurrence, assigning one occurrence per profile if at least one hydrometeor was detected in that profile. In addition, right panels shows the frequency profiles through the entire dataset for each season.

Notably, monthly evolution of ice and liquid hydrometeors keeps aligned with temperature and relative humidity profiles. For both cases, higher frequencies were encontered in the end of cold regimes (November) and during warm regimes (Dec-May). Where we can easily find frequencys ranging 20% for liquid and 15% for ice. Also, a proeminent minimun occured exactly in summer, which can be better visualized in the right panels of Figure 3. These seasonal profiles showed an increase for the liquid case from summer to winter, and for the ice case, an increase from summer to fall, keeps almost the same in winter, and reach a maximun in spring. This feature is strong for the liquid case, where theres no diference between winter and spring. Also relevant frequencies for this case is below 2 km. On the contrary, middle panel of Figure 3 shows that ice particles have a spread distribution on the vertical, overcomming 10 km with slight maximas in 2 km and 8 km.

Therefore, by converting hydrometeors into cloud types using the cloud classification algorithm (refer to section 2.3), Figure 4) illustrates the frequency of cloud availability per month in the top panel, and the frequency of occurrence of single-layer clouds for different cloud types relative to the available data in the bottom panel. It's important to note that each cloud type in the legends are single-layer clouds, with any multilayer clouds considered as separate case. Indeed, all subsequent analyses

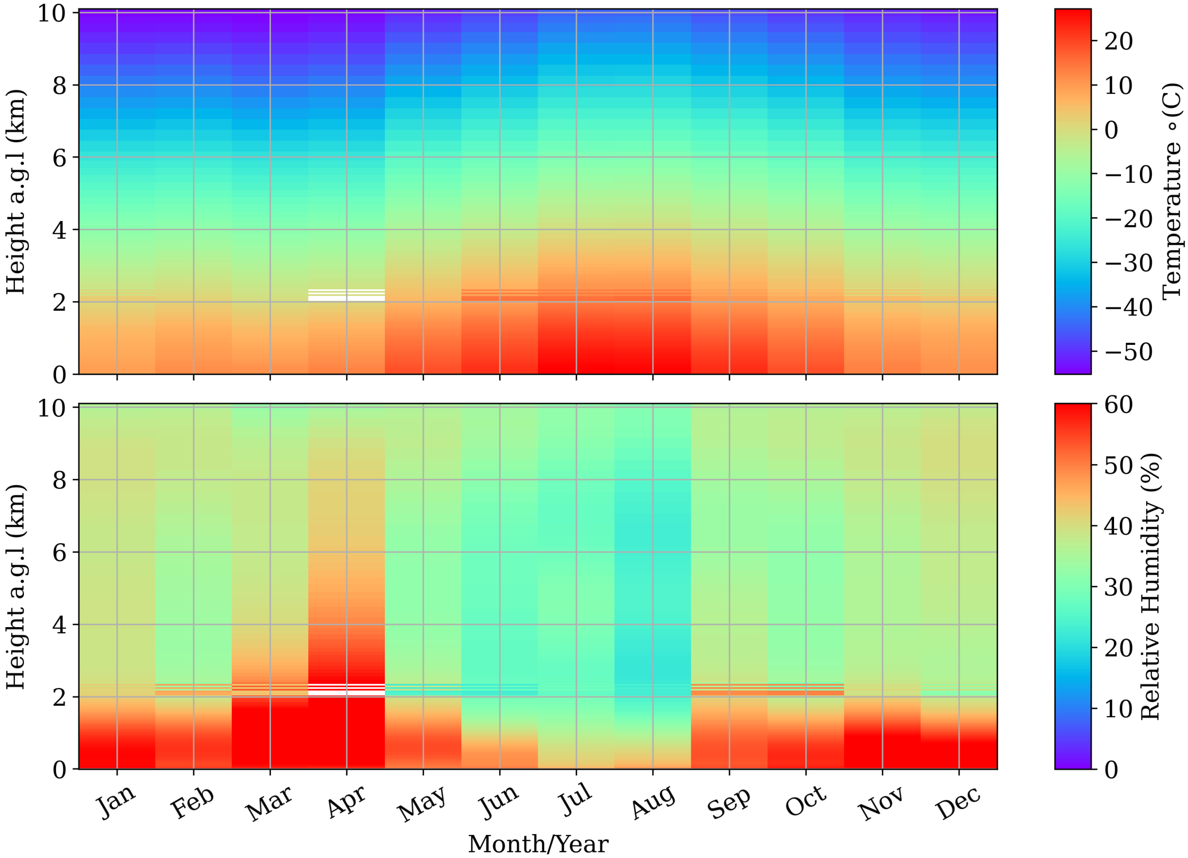


Figure 2. Montlhy evolution of averange temperature and humidity profiles obtained with the microwave radiometer.

focus on single-layer clouds, as the CLOUDNET target classification product can be highly inaccurate in these cases due to errors in LWP partitioning between layers (Nomokonova et al., 2019).

In principle, the top panel displays a significant amount of missing data and no-cloud periods, with minimal presence of liquid-precipitating clouds and a predominant occurrence of precipitating mixed-phase clouds. Similarly, the bottom panel offers a more direct comparison of single-layer clouds, revealing that liquid-precipitating mixed-phase clouds are dominant in cold regimes (winter - spring) and absent in summer (June - August). Their occurrence peaks at 23% in winter and 30% in spring. Moreover, ice clouds are the most frequent toward the end of summer and the beginning of fall. Overall, each cloud type exhibited a similar trend, with fewer occurrences in warm regimes and higher occurrences in cold regimes. However, only precipitating mixed-phase clouds displayed significant seasonal variability, with almost zero occurrences in summer. Generally, there are no rain events in this season.

Although precipitation-mixed-phase clouds are the most frequent, we excluded all precipitating cases from the subsequent analysis. Precipitation can heavily attenuate reflectivity and introduce bias in the mean Doppler velocity due to aliasing erros.

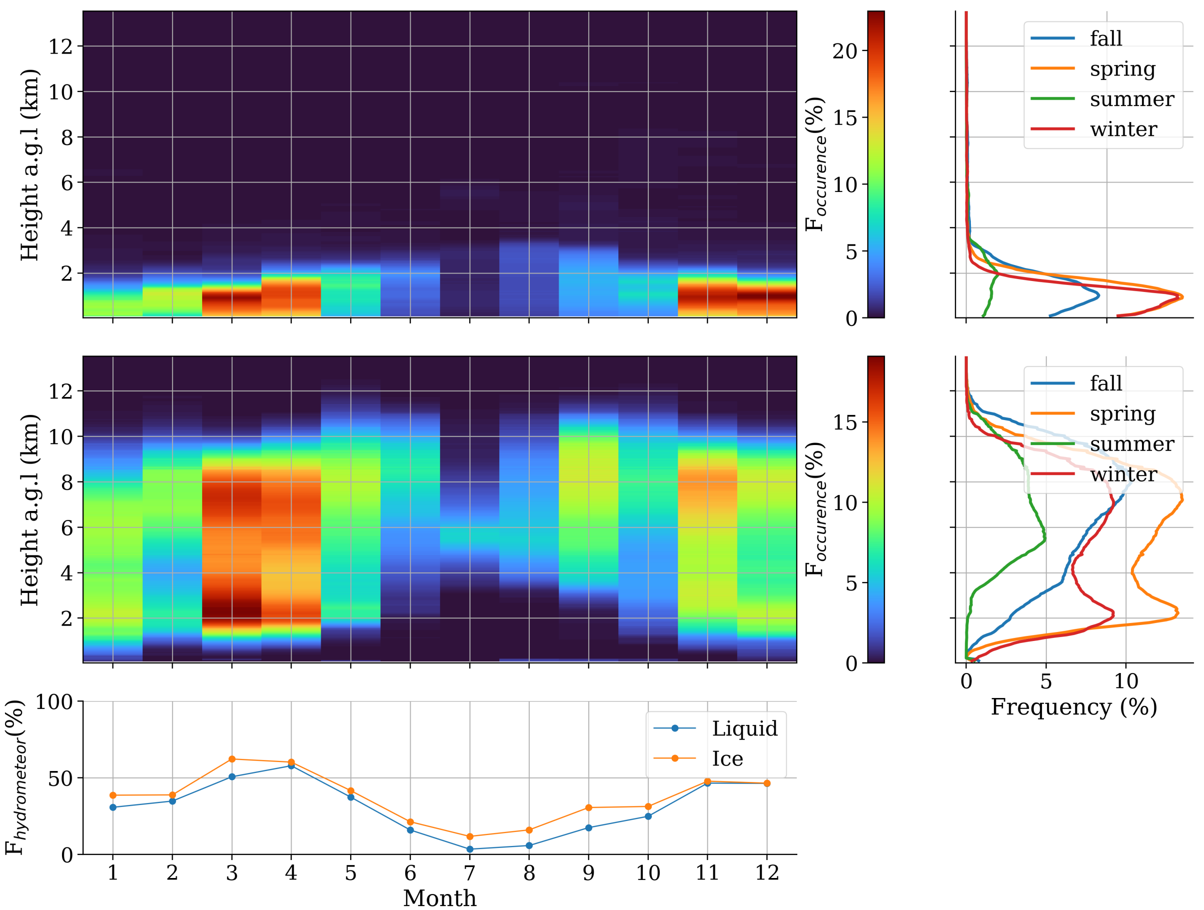


Figure 3. ...

Additionally, it can completely attenuate the ceilometer backscattering signal. Given that the CLOUDNET target classification relies on these variables, this product can be significantly affected. Since the impact of this effect on hydrometeor classification is still not well understood, we opted not to analyze this case within the context of the study.

In Figure 5, a comparison of cloud thickness (ΔZ), base, and top for each cloud type by season is presented. This figure displays a boxplot for cloud thickness for each season within a density distribution contour. Consistent with previous findings, the top panel confirms that mixed-phase clouds exhibit the largest and most variable cloud thickness at our station, followed by ice clouds and then liquid clouds. Their median values are 2.0 (XXX) m, 1.2 (XXX) m, and 0.2 (XXX) m, respectively. However, it's noteworthy that the most likely values are consistently smaller than the median (as indicated in the top panel of Figure 5). This can be clearly seen for ice and mixed clouds, and is also observed for liquid clouds, although it might not be readily apparent in the graph due to the scale of the y-axis. On the other hand, no significant seasonal trend was observed for this variable, although it was clearly observed for cloud base and top.

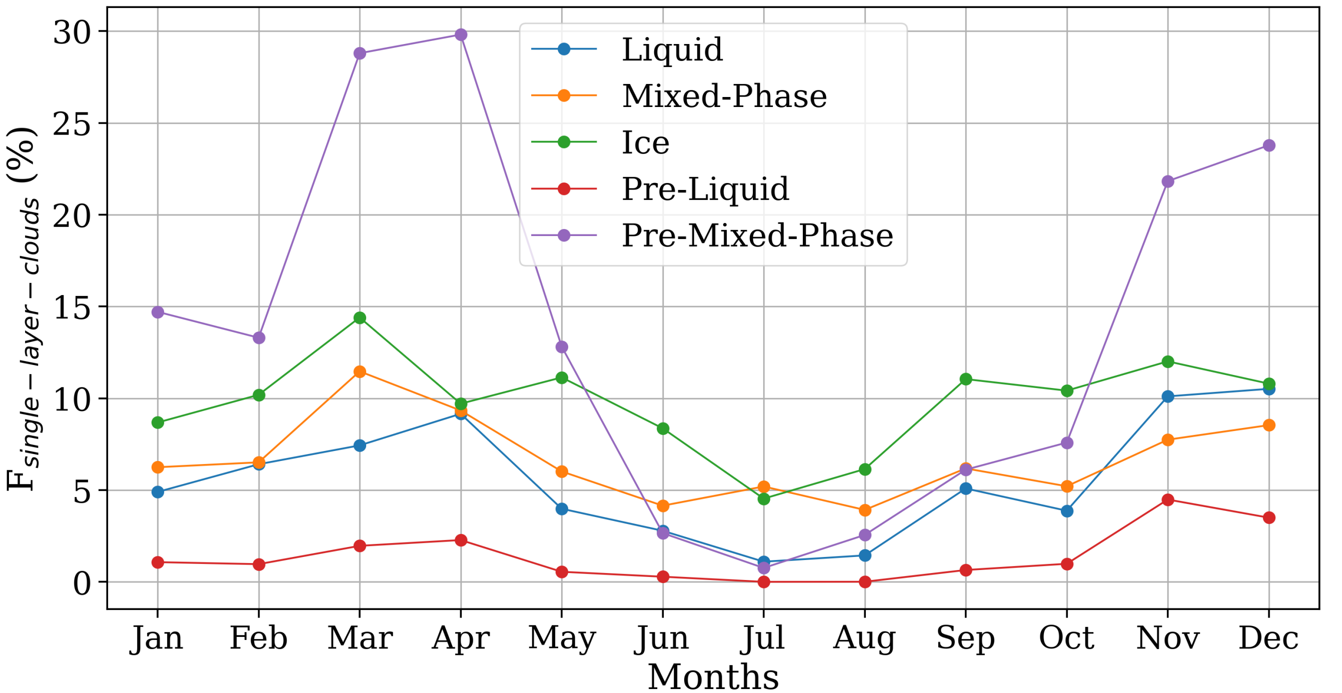


Figure 4. ....

In hot regimes (summer and fall), cloud base tends to reach higher levels as can be seen by the middle panel of Figure 5. Especially in summer, a prominent mode appears around 5 km, even for liquid clouds that usually do not reach this level for the rest of the year. Indeed, most dramatic changes in the vertical distribution of the cloud base happened in summer, principally for liquid clouds. This clouds changed from sharp distribution at low levels to a spread one in summer, extending up to 5 km as mentioned. Still, in a general framework, cloud bases are typically higher for ice clouds, followed by mixed clouds and then liquid clouds. Also, except for summer, cloud base for mixed phase clouds presents the largest vertical variability, ice cloud bases are normally found above 6 km, and liquid cloud bases stay below 2 km. Consequently, the same qualitative behavior happened for cloud tops, once no significant changes in cloud thickness was observed before. Additionally, since liquid clouds are very thin, just a slight difference between cloud base and top distribution has been observed.

4 Clud properties evaluation

Next, allocation of LWP frequencies can manage to discern liquid hydrometeors partitioning between clouds. These frequencies are shown within histograms in top-left panel of Figure 6 (again, discarding rain events and multi-layer clouds), where, both liquid and mixed-phase clouds appear to contain similar amounts of water. This agreement between both cloud types can be assigned to the assortment algorithm design. As explained in section 2.3, this algorithm identify clouds for indiviadual profiles

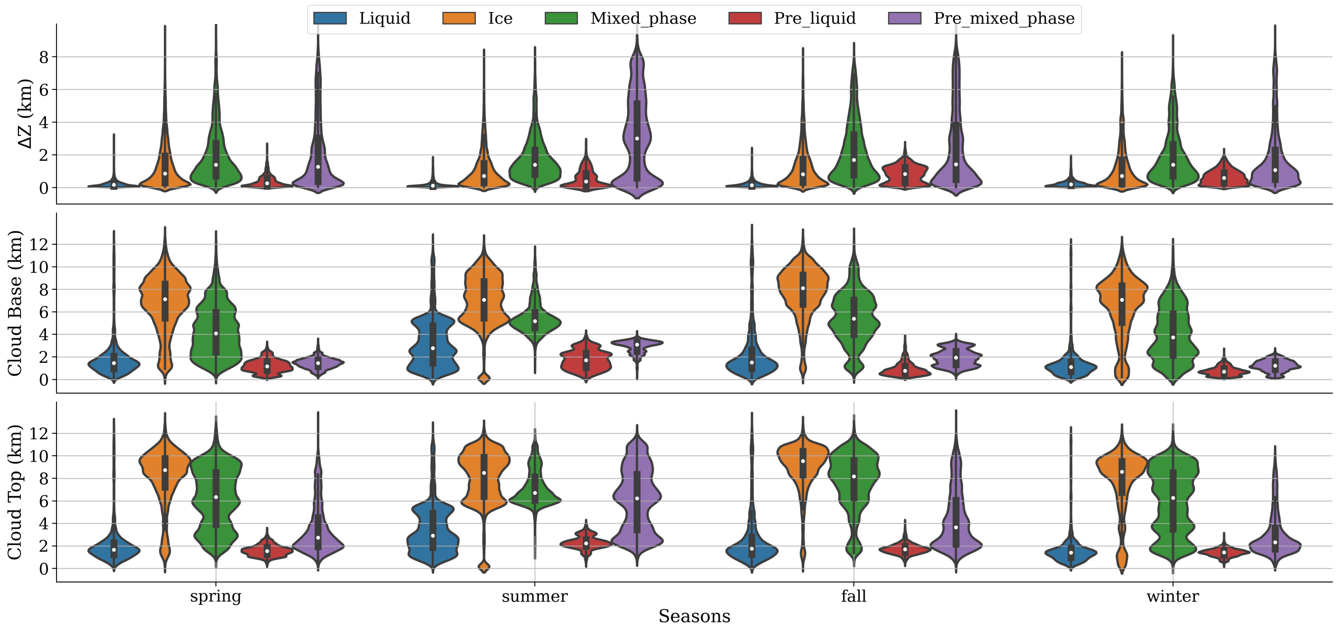


Figure 5. ....

of 30 s without considering a continuos sequence of same cloud profiles. Then it is possible to assign profiles of liquid clouds among mixed phase cloud, while they still being part of the same cloud as a whole. Nevertheless, these clouds presented different montlhy distributions of LWP. Top-right panel of Figure 6 shows the montlhy median values of LWP in solid lines, and the 25 - 75 percentiles intervals throughout the painted area.

The horizontl grid lines indicate the bins of the histogram on the left. Therefore, for mixed phase clouds, the median of LWP are ranging within the most plausible ranges of 0-10 g m^{-2} (55%XXX of occurence) and 10-20 m^{-2} (20%XXX of occurence) throughout the entire year. Alternatively, median values of LWP for liquid clouds surpass this interval in Feb-May, hiting up to 30-40 g m^{-2} in February, with only 6% of frequency of occurence. Above this ranges, the frequency of occurence decrease almost exponentially. Also, months with more reliable statistics due to data availability are shown in the botom-right panel of Figure 6, which are the months in colder regimes. On the whole, a slight seasonal pattern of LWP was found for liquid clouds, with higher values and large variability during spring and fall, and low values during summer. For mixed phase clouds, no seasonal pattern of LWP was found, values normally range between 0-20 g m^{-2}.

Same analises was done for ice hydrometeors, which can be partitioned between ice and mixed phase clouds. Left panel of Figure 7 also shows same feature between these cloud types, whose explanation is the same as the liquid case. However, IWP for mixed phase cloud is sistematicaly higher than those for ice clouds. Also, its montlhy distributions shown in top-right panel of Figure 7 suggests that median values of IWP for ice clouds is whithin the interval 0-10 g m^{-2} (60% of occurence). Again, note that in this figure, horizontal grid lines indicate bins of the left side histogram. For mixed phase clouds, IWP median values usually remains whithin 0-20 m^{-2}, surpassing it only in Aug-Oct. In these months, the median range between 20-50 g m^{-2},

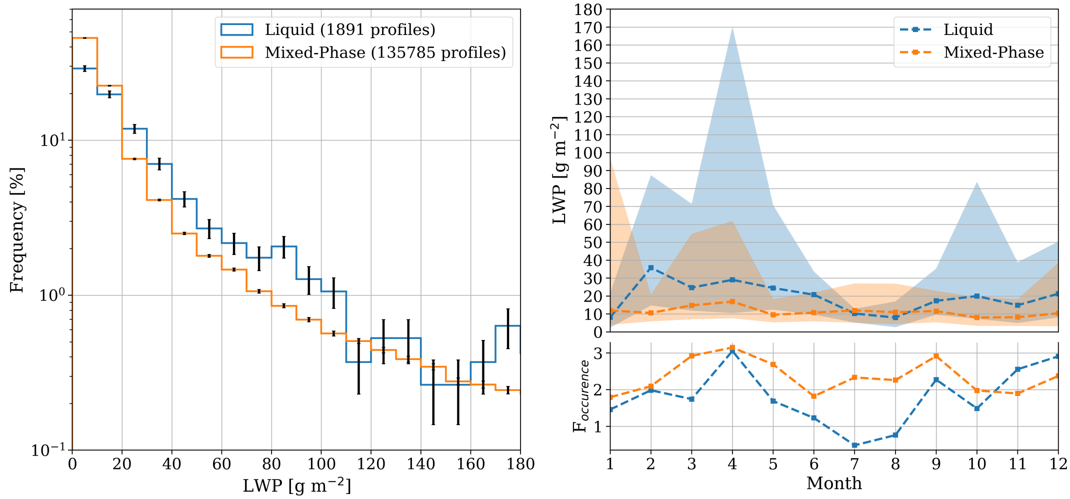


Figure 6. ....

with roughly less than 6% of occurence. Again, higher values are unlikely and frequency of occurence drecease exponentialy for both cloud type. Additionally, no seasonal pattern was identifyed and IWP for mixed phase clouds have higher variability compared with ice clouds.

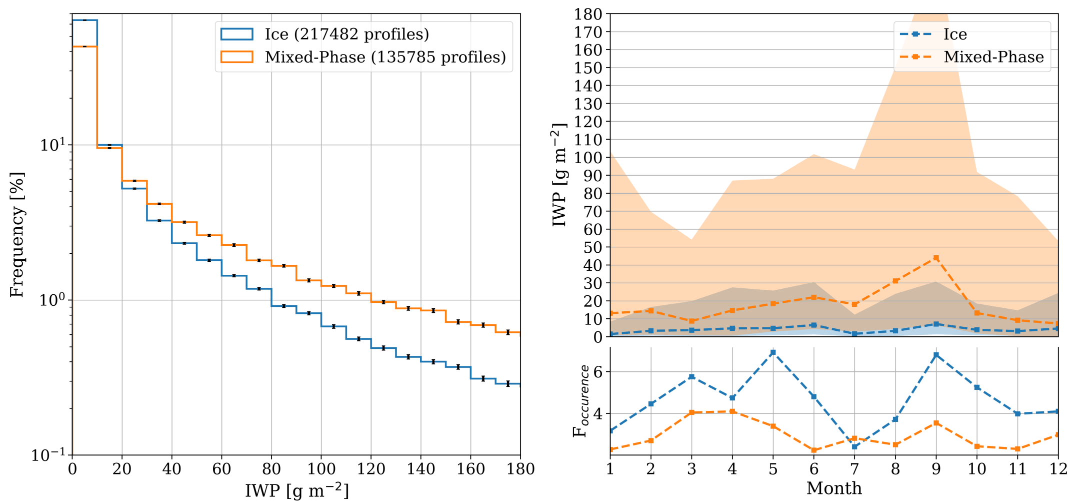


Figure 7. ....

Figure 8 shows the monthly and seasonal mean vertical distribution of liquid water content (LWC) within liquid (top panel) and mixed-phase clouds (bottom panel). In liquid clouds, mean values can reach 0.1 g m $^{-3}$ until approximately 2.5 km above mean sea level (equivalent to 1.2 km in height above ground level), as expected (see figure 3). However, seasonal variations are indistinguishable due to the significant variability of this variable. It's worth noting that the shaded area in the seasonal profiles on the left panel represents the standard deviation divided by 10. Moreover, the monthly time evolution indicates that LWC is lower in July, during the summer, consistent with the observations of liquid water path (LWP) (refer to Figure 6). Additionally, during this month, elevated values in a yellowish tone are dispersed vertically up to 5 km. The vertical positioning of the yellowish tone corresponds to the most frequently observed cloud base height values depicted in the middle panel of Figure 5.

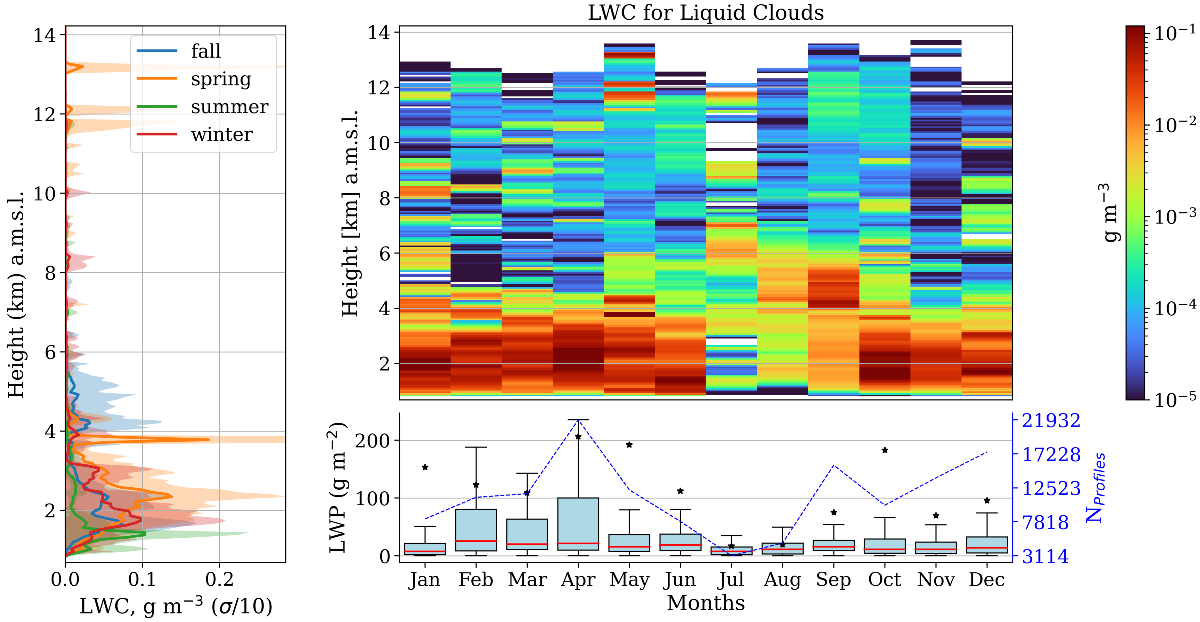


Figure 8. ....

Mixed-phase clouds depicted in Figure 9 display a notably broader distribution of liquid water content (LWC), despite exhibiting smaller values compared to liquid clouds. Interestingly, monthly profiles of LWC for colder periods (spring and winter) showed fewer occurrences of high values than warmer periods (summer-fall) below 5 km. However, as the liquid case, distinguishing seasonal profiles is challenging due to significant variability. Additionally, high values of LWC are observed at higher levels during summer, coinciding with the features of cloud base and top height observed in Figure 5.

Regarding ice water content (IWC), almost the same monthly mean values were found for ice (Figure 10)and mixed phase clouds (Figure 11), in addition to the same mean vertical distribution by season. This outcome was unexpected given our previous demonstration that mixed-phase clouds exhibit higher values of ice water path (IWP) than ice clouds. Consequently, the monthly distribution of IWC appears more asymmetric than that of LWC, as this behavior is not observed for LWC. Consequently, the most probable value of IWC each month significantly differs from the average. Therefore, the monthly average proves to be an inadequate parameter for representing IWC. Moreover, a more accurate portrayal of IWC would entail sharper and higher distributions, as depicted in Figure 5, which cannot be achieved through averaging. Conversely, seasonal

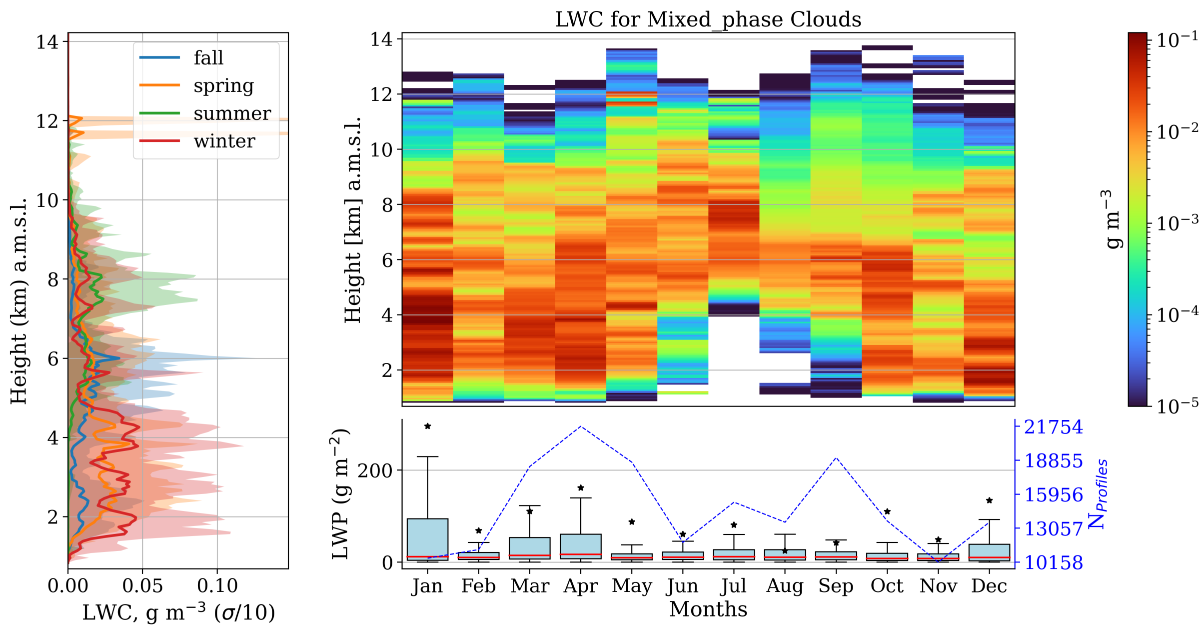


Figure 9. ....

patterns can be discerned, with maximums around 0.14 and 0.19 g m $^{-3}$ for ice clouds in cold and warm regimes, respectively. For mixed-phase clouds, these values are 0.12 and 0.19 g m $^{-3}$.

To compare cloud microphysics between ice and mixed-phase clouds, droplet effective radius of Figure 12 and 13 are presented using the same color scale. Left panel of this figure shows the seasonal vertical distribution of the median droplet effective radius where shaded area represents the 25 and 75 percentiles. Right top panel shows the monthly vertical distribution of the median, and bottom panel shows the median (solid line) and the distribution of R_{eff} by month. Additionally, this panel has second y axis with the number of profiles used each month to compute the medians on the above panel. In the case of liquid clouds, the left panel of Figure 12.a indicates no significant seasonal differences, with values ranging from 5 to 8 μm. Monthly distribution of R_{eff} (right-top panel of figure 12) show higher values in light blue tones, approximately around 10 μm between 6 and 8 km altitude, and only a few layers displaying green tones, indicating particles with a size of 15 μm. Additionally, monthly examination of violin plots of R_{eff} in the bottom panel demonstrates variations in medians and modes between 4.5 and 6.5 μm, corresponding to the dark blue tones below 4 km altitude in the above panel, consistent with the presence of large values of LWC in Figure 8

Concerning mixed-phase clouds, Figure 13 presents a clearer monthly distribution. Although the left panel indicates higher values for liquid droplet radius compared to liquid clouds, no significant seasonal differences were observed. However, it is evident that higher values are encountered between 4 and 6 km of altitude, and them could vary between 13 and 30 μm with peaks during summer (as observed in the top-right panel). Additionally, the violin plots (bottom-right panel) illustrate a higher probability of reaching higher values of R_{eff} during summer. Nevertheless, it's noteworthy that particularly for colder periods

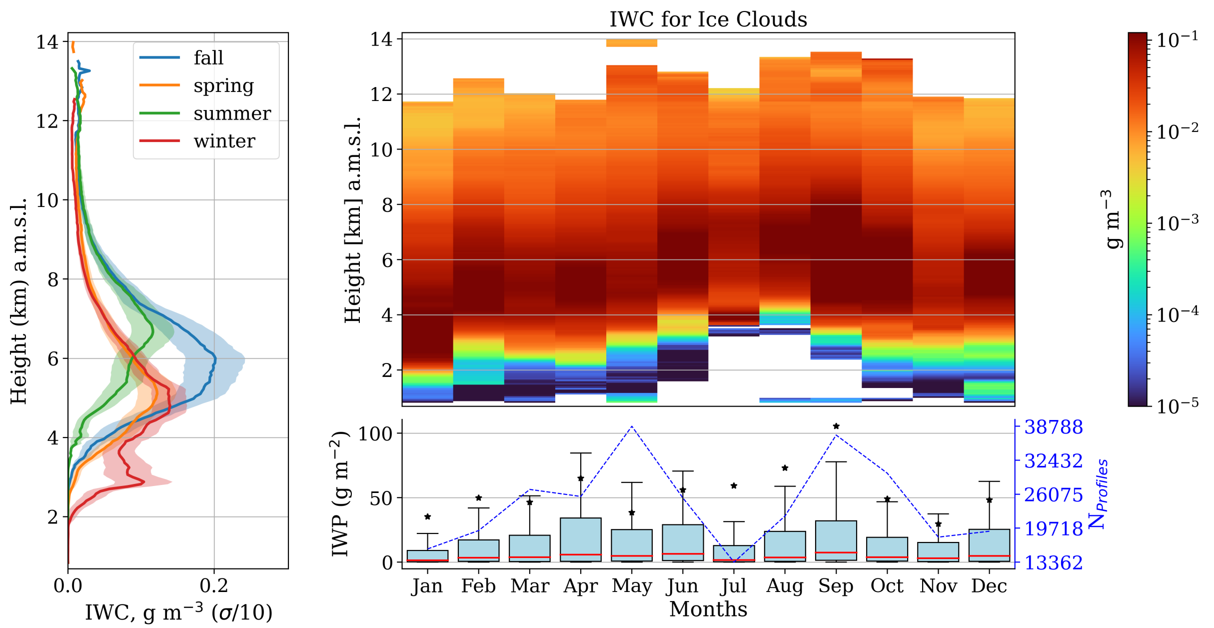
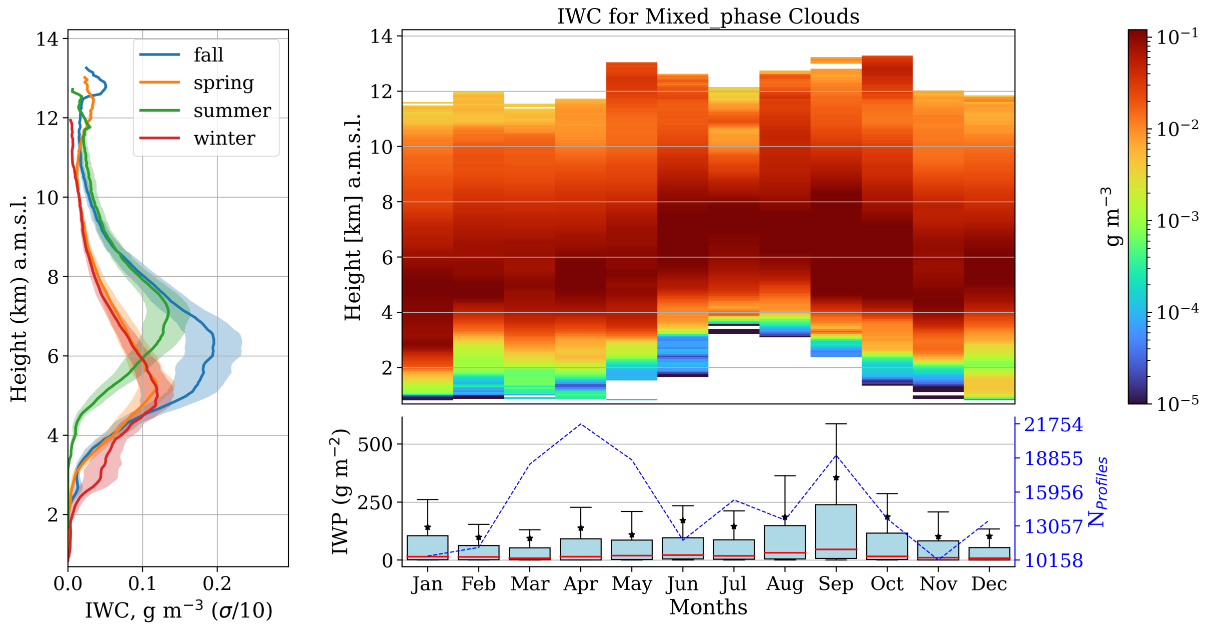


Figure 10. ....



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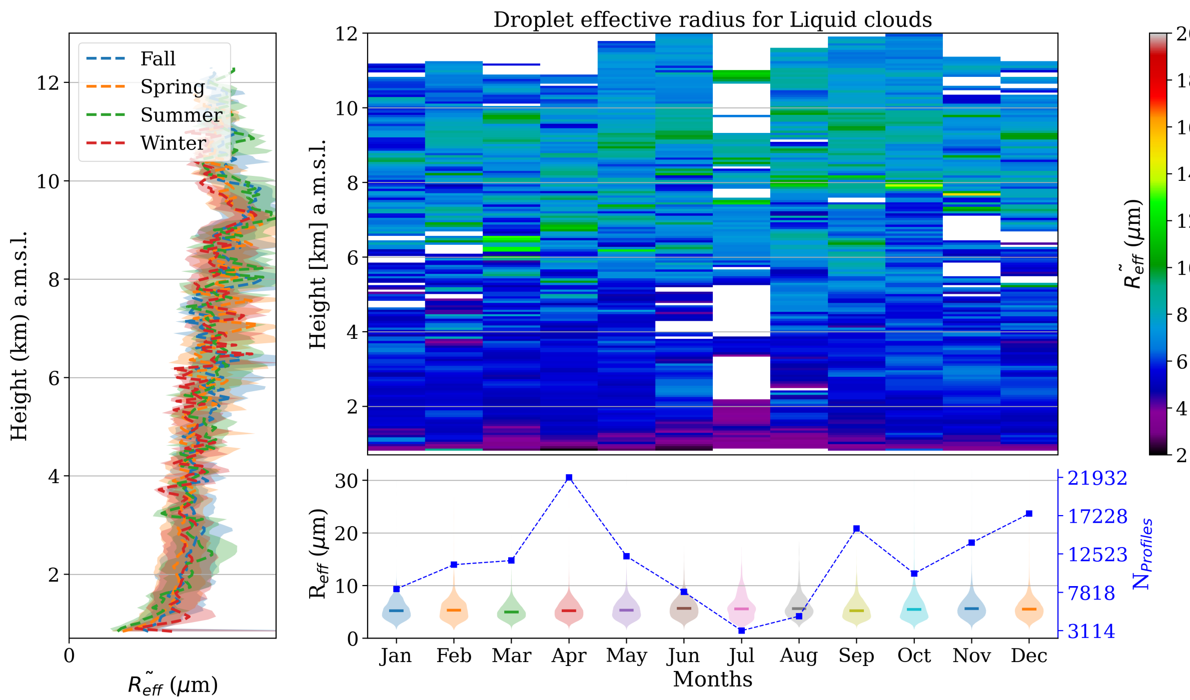

Figure 12. ....

(December to May), the median and mode of R_{eff} are approximately 12 and 5 μm, respectively. These values correspond to a range of colors from dark blue to light green in the above panel, predominantly found below 4 km and between 6.5-10 km altitude.

Repeatedly, the left panel of Figure 14 shows no significant seasonal variations in the median ice effective radius for ice clouds. However, higher values of effective radius were notably observed during summer (June-August), reaching up to 65 μm (as shown in the top-right panel). This aligns with the distribution of summer R_{eff} depicted in the bottom-right panel, where median values exceed 40 μm in July and August. Additionally, these months exhibit sharper distributions of R_{eff}. Overall, the R_{eff} fluctuates around 35 μm for other seasons, with modes around 20 and 30 μm. This corresponds to colors ranging from light blue to green tones in the above panel, indicating that the most common Reff for ice clouds is typically situated between 8 and 12 km altitude (above mean sea level), likely associated with cirrus clouds. Conversely, while higher R_{eff} values may be found in low to middle levels, their probability of occurrence is minimal, as evidenced by the small tails of each distribution extending beyond 60 μm.

On the other hand, the median vertical distribution of R_{eff} for mixed phase clouds (left panel of figure 15) exhibits a significant difference between cold (winter-spring) and hot periods (summer-fall) between 8 - 11 km. In addition, summer presented high R_{eff} within 2-4 km, which is also observed in the monthly vertical distribution on the right-top panel. However, no clear differences between these months were observed in the monthly distribution of R_{eff} in the right-bottom panel, except

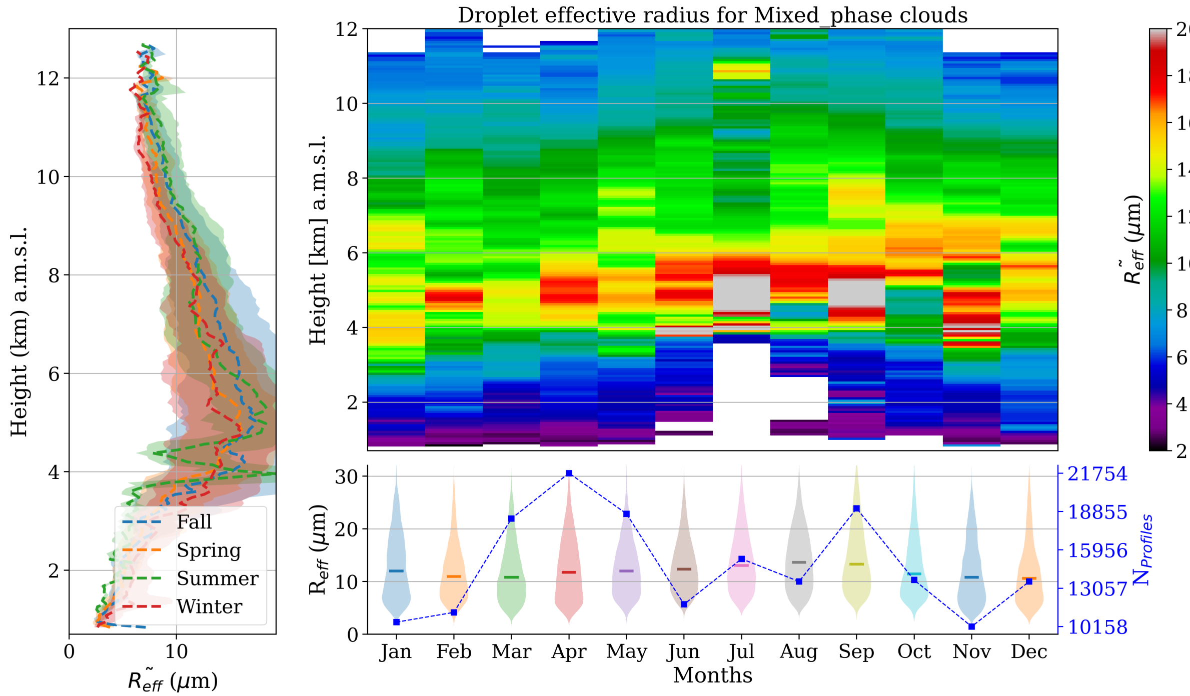

Figure 13. ....

that July has a slight sharper distribution. This panel also shows that median values are normally ranging around 45 μm. It is worth to note that differently from previous cases, the modes are above the median, ranging around 50 μm. Therefore, the most probable R_{eff} in the above panel corresponds to the band around the yellow tone, i.e. approximately between 2 to 8 km. During summer this layer is moved upwards as also seen for ice clouds.

In summary, liquid clouds exhibit slight larger integrated water path (LWP) compared to mixed-phase clouds, with typical values ranging between 0 and 20 g m^{-2} for both, occasionally exceeding 40 g m^{-2}. However, liquid clouds can potentially reach 30 g m^{-2}, whereas mixed-phase clouds cannot. Additionally, the monthly evolution of this parameter does not display a strong seasonal pattern for both types of clouds, consistent with their frequency of occurrence (see Figure 6 and 7). Notably, the only cloud type showing a pronounced seasonal trend is the liquid-precipitating mixed-phase clouds, indicating that precipitation in Granada is mainly associated with mixed-phase clouds. Regarding ice water content (IWC), its values are consistently greater for mixed-phase clouds, which contain the majority of ice hydrometeors in the atmosphere.

Similarly, no significant seasonal differences in liquid water content (LWC) profiles were observed for non-precipitating clouds. However, liquid clouds predominantly exhibit LWC below 3 km, with higher values in summer, contrasting with more evenly distributed LWC values in mixed-phase clouds. The same behavior was observed for droplet effective radius, despite the fact that the radius tends to be larger for mixed-phase clouds. In terms of IWC, its values are vertically distributed for both ice and mixed-phase clouds. However, the most likely values for ice clouds are between 8 and 11 km, whereas for mixed-phase

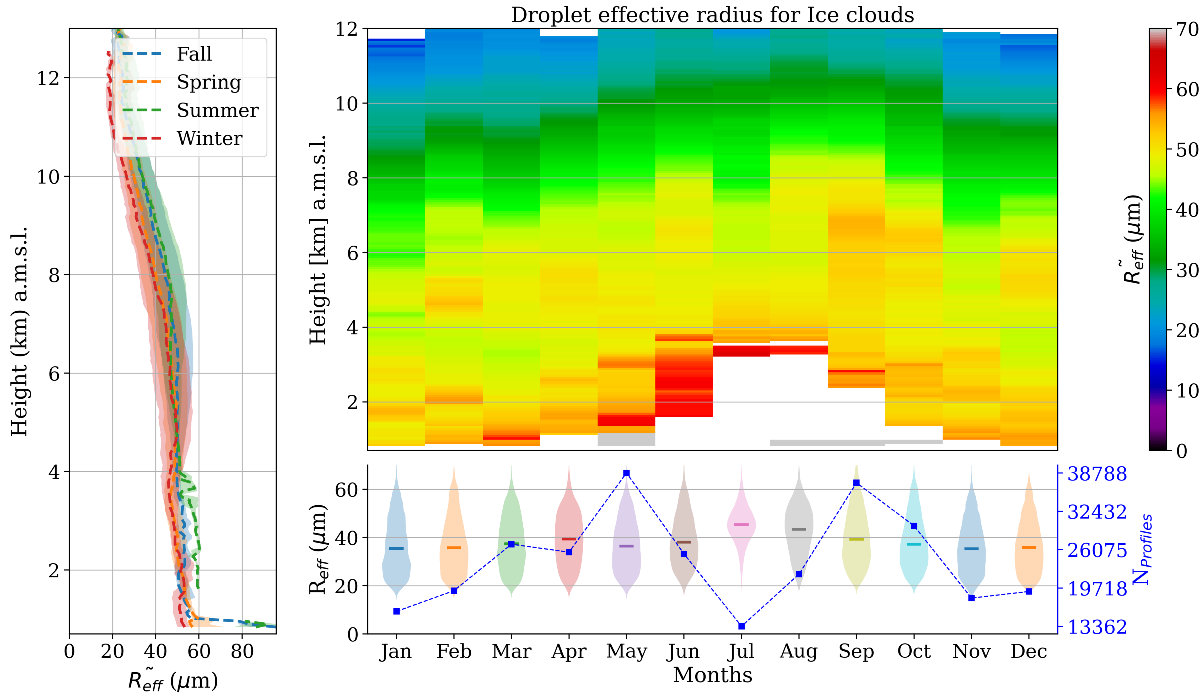

Figure 14. ....

clouds, they range between 2 and 8 km. Additionally, similar behavior was found for ice effective radius. Subsequently, mixed-phase clouds exhibit greater cloud thickness compared to liquid clouds, while no discernible seasonal trend was identified for this parameter. Nevertheless, clear seasonality was observed for cloud base and top, which were higher in warmer periods and lower in colder ones, as expected. However, during summer, unusual vertical distributions were observed, with liquid clouds particularly affected, reaching altitudes of up to 5 km.

5 Summary and conclusions

The study of cloud coverage and its physical properties are the first step in understanding cloud radiative effects. In order to retrieve these variables, CLOUDNET post-processed data of FMCW 94 GHz cloud Doppler radar, HATPRO radiometer and CHM15k ceilometer were used. Additionally, complementary analysis of thermodynamic conditions was performed with the HATPRO radiometer vertical profiles. Temperature and relative humidity profiles tracked cold (fall-winter) and warm regimes (spring-summer). The assortment algorithm was applied using the CLOUDNET target classification product to classify different single layer cloud types. Inter-annual observation on cloud occurrence revealed a seasonal trend, where mixed phase clouds are the prevailing species, followed by ice clouds, especially during spring and winter. In these seasons, precipitating mixed phase clouds can reach 30%XXX and 25%XXX of frequency, respectively, and ice clouds 15%XXX and 12%XXX.

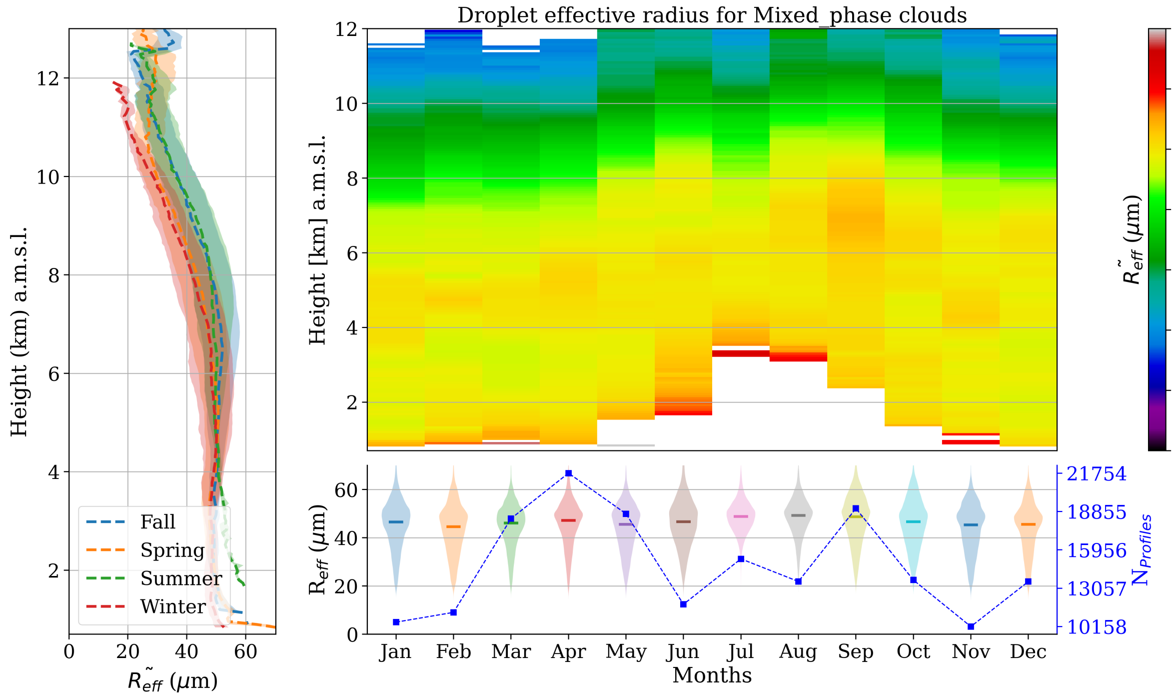

Figure 15. ....

Other cloud types have no strong seasonal pattern, and precipitating liquid clouds are absent throughout the year. In addition, the total single layer cloud frequency in summer is less than 8%. Mixed-phase clouds exhibit the thickest layers, followed by ice clouds and then liquid clouds. The median cloud thickness for these clouds is 2.0XXX, 1.2XXX, and 0.2XXX, respectively. However, due to their considerable variability, no distinct seasonal pattern was discernible for this property. Consequently, thicker layers of mixed-phase clouds contain more ice content than ice clouds layers. Although typical values for both are 0-20 g m^{-2}, mixed phase clouds have systematically higher IWP, with an expressive increase in fall, reaching 20-50 g m^{-2}. Conversely, liquid clouds contain slightly more liquid content than mixed-phase clouds, despite being thin. This indicates that liquid layers consistently maintain thinness, even within mixed-phase clouds. In fact, they have almost the same amount of water, within the interval of 0-20 g m^{-2}, but liquid clouds have higher probability to reach higher values, specially in spring and fall (see figure 6).

LWC and droplets effective radius does not show a clear seasonal trend for mixed and liquid phase clouds. Effective radius has almost the same distribution along year, with a median of 5 μm for liquid clouds, and is found mainly below 3 km. Differently mixed phase clouds have a radius that can vary a lot from the surface to 10 km, but in general, most frequent values between 5 and 12 μm are found between 1-2 km and 5-9 km. Concerning LWC, despite the fact that it was easy to identify higher values in cold regimes and smaller ones during summer, the variability of data goes through several orders of magnitude. It was observed for liquid and mixed phase clouds.

With respect to ice and mixed phase clouds, the same difficulty to reach a seasonal pattern for IWC was identified. Month profiles did not show a significant difference, except that in summer, values are shifted upwards in the atmosphere. For ice clouds, effective radius can reach 65 μm in summer, but normally they vary between 20-30 μm around 8-12 km. For mixed phase clouds, ice effective radius fluctuates around 50 μm between 1-7 km.

Concerning cloud base height, a significantly seasonal difference on vertical distribution was observed in summer. Normally, the median value of ice cloud base height is 8 kmXXX in fall and 7 kmXXX for the rest of the year. For mixed phase, median values are 4 km and 5 km in cold and warm regimes, respectively. However, an unnusual mode around 5 km appeared in summer, especially for mixed phase clouds, which also present a sharper distribution compared to other seasons. Nonetheless, in this season, liquid clouds presented a spreader distribution, with a median of 3 kmXXX, against a median of 1 kmXXX in other seasons. Cloud top height followed a simmilar behavior. In colder regimes (winter-spring), cloud top is 1.5 kmXXX, 9 kmXXX, 6 kmXXX, for liquid, ice and mixed clouds, respectively. During warm periods (summer-fall), medians changed to 2.5 kmXXX, 8.5 kmXXX, 7.5 kmXXX. However, only in summer, the meadian cloud base height for liquid clouds were 3 kmXXX. Finally, a comprehensive analysis of cloud occurrence and physical properties was conducted at the ACTRIS-CCRESS Granada station. Notably, this is the first study on cloud properties in the southeast of Spain, an area recognized as a hotspot for climate variability. Consequently, it has the potential to enhance our understanding of global cloud properties, as the methods employed here can be applied to other stations as well.

Acknowledgements. TEXT

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